because the grouping eliminates noise (better shows a pattern) and provides a balanced data set (e.g., the underlying sample data is not skewed to certain sizes). When we group the data, we filter out small sample categories. For example, require at least 10 or 20 data points for each grouping bucket to be included in the estimation analysis.

Next, run a regression of the average cost of each POV category as a function of the POV rate. That is:

$$Avg\ Cost = \alpha_0 + \alpha_1 \cdot POV$$

Solve for α_0 and α_1 via linear regression analysis. Then estimate b_1 from the estimated alphas as follows:

$$\hat{b}_1 = \frac{\alpha_1}{\alpha_0 + \alpha_1}$$

Notice here that the intercept term α_0 corresponds to POV = 0. It further indicates the cost quantity that will be incurred regardless of the POV rate in the analysis[2].

Step 2: Estimate a_i Parameters

The process to estimate parameters a_1, a_2, a_3 follows from our estimated value for $\hat{b}_1$ above. We have:

$$MI = \hat{b}_1 \cdot I^* \cdot POV + (1 - \hat{b}_1) \cdot I^*$$

Next factor the equation as follows:

$$MI = I^* \cdot \left(\hat{b}_1 POV + \left(1 - \hat{b}_1\right)\right)$$

Dividing both sides by $\hat{b}_1 POV + \left(1 - \hat{b}_1\right)$ gives:

$$\frac{MI}{\left(\hat{b}_1 POV + \left(1 - \hat{b}_1\right)\right)} = I^*$$

[2]During my economic consulting days with R.J. Rudden Associates, Inc., we used a similar analysis called a zero mains study. This study served to determine the breakdown between the fixed cost of installing a gas main and the variable cost of the gas main that is dependent upon the size of the main. The intercept term is equivalent to a gas main of zero inches and represents the fixed cost. The two-step regression analysis follows from this type of study.